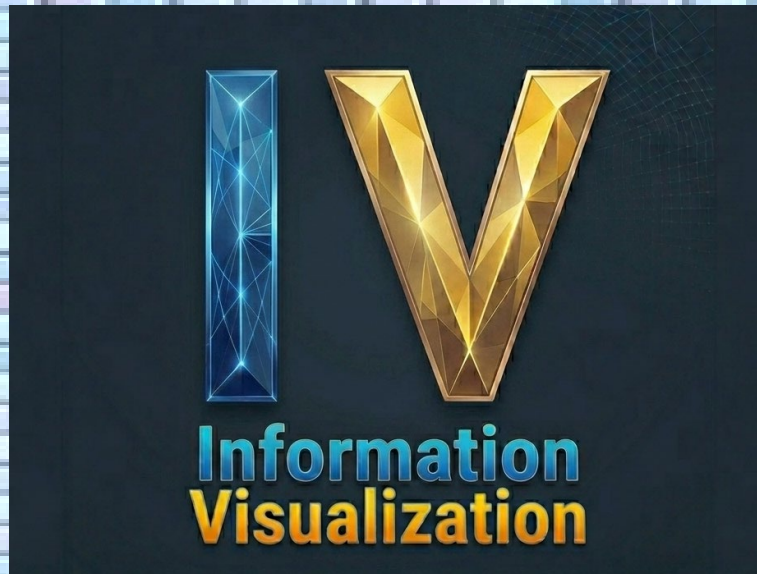


Vorlesung
Informationsvisualisierung

Kapitel 2 INFOGRAFIKEN



Lecture
Information Visualization

Chapter 2 INFOGRAPHICS

First Tutorial

2

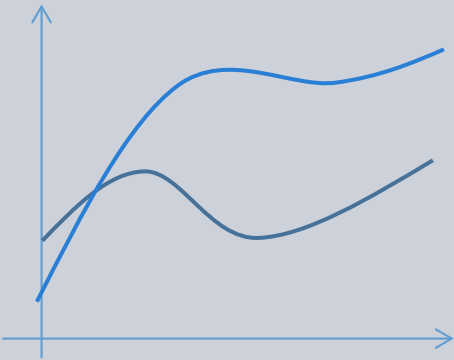
- Tuesday, **28.04.2026**, 10-12 in HZ 203
 - 1st and 2nd exercise sheet

Chapter
2.1

Diagrams

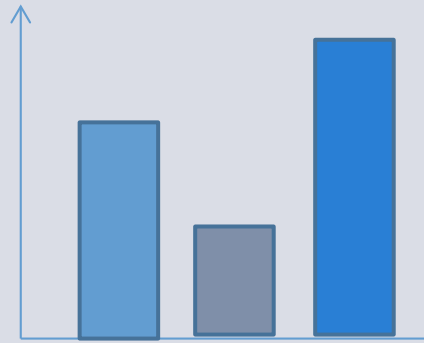
Commonly used Diagrams

4



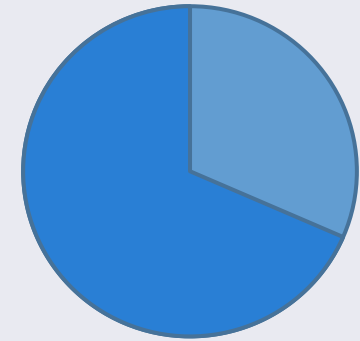
Line charts

- e.g. temporal sequence



Bar charts

- e.g. frequencies

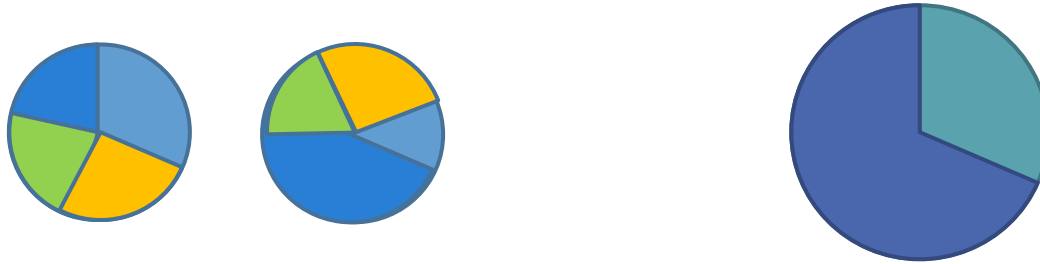


Pie charts

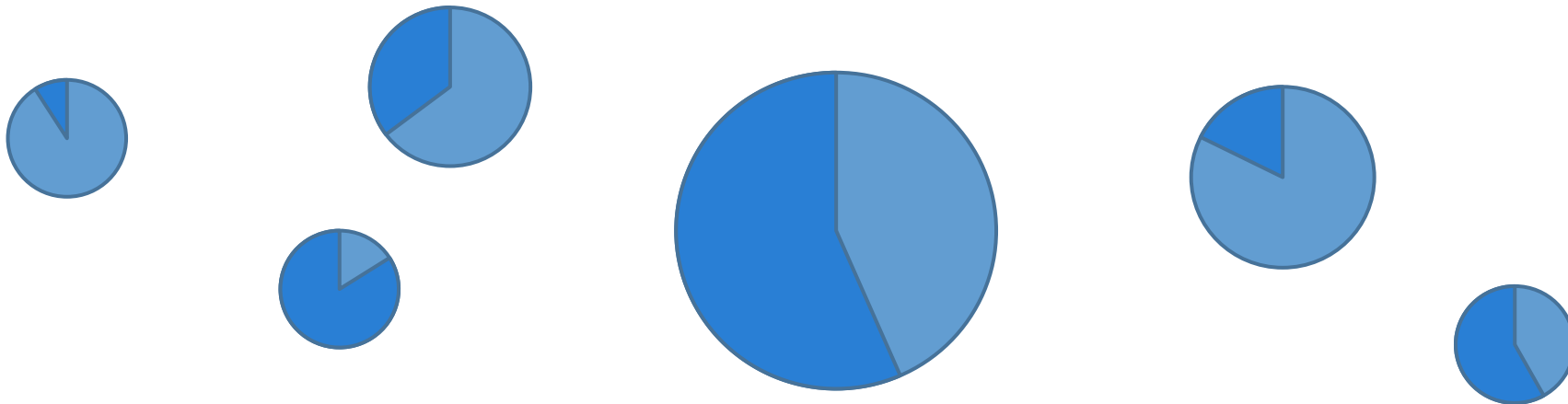
- e.g. categorial distribution

Controversy: Pie charts

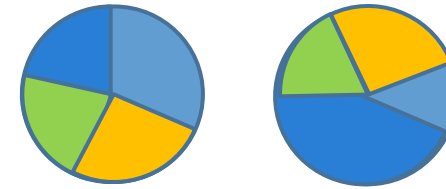
5



“the only thing worse than a pie chart is several of them”
[Tufte, 1992]



Pi charts vs. Bar charts



6

Pro pie chart

- Works well for comparing the size of a single category with the total size.



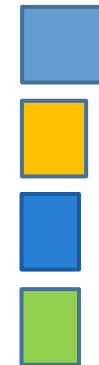
Pie chart

Contra pie chart

- Difficult to compare size of different categories.



Stacked bar chart



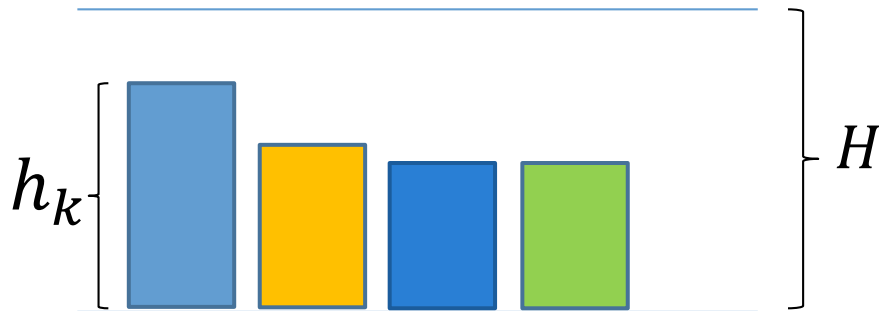
Bar chart

Draw chart for values $v_1, \dots, v_n$

7

Bar chart

$$h_k = \frac{v_k}{\max_{i=1}^n v_i} * H$$



Pie chart

$$\alpha_k = \frac{v_k}{\sum_{i=1}^n v_i} * 2\pi$$



8

[illegible]

Das Diagramm stellt das Spiralmodell nach Boehm dar. Es ist in vier Quadranten unterteilt, die die Phasen des Modells repräsentieren:

- Oben links:** Festlegung der Ziele
- Oben rechts:** Vergleich alternativer Lösungsmöglichkeiten
- Unten rechts:** Entwicklung und Test
- Unten links:** Planung des nächsten Zyklus

Die Spirale selbst ist in vier Segmente unterteilt, die die Prototypen darstellen:

- Prototyp 1:** Innerster Kern der Spirale.
- Prototyp 2:** Zweites Segment der Spirale.
- Prototyp 3:** Drittes Segment der Spirale.
- Prototyp 4:** Viertes Segment der Spirale, das am weitesten rechts liegt.

Ein Pfeil am unteren Rand zeigt den Fortschritt der Spirale an.

8

The screenshot shows the 'The Wall of Debt' app interface on an iPad. The main title is 'The Wall of Debt'. Below the title, there's a red brick icon and the text '1 Brick = £1m'. A large red bar chart represents the national debt at £1059 billion. To the right, there are two smaller bar charts: one for GDP at £23 billion and one for Income at £116 billion. The app has a 'PLAY' button at the bottom left and a '2016, V2' label at the bottom right.

Animated timeline

Sparklines

9

A **sparkline** is a small intense, simple, word-sized graphic with typographic resolution. [Tufte, 2004]



Last 12 months of the U.S. unemployment rate, which rose to 9% in April. More data: <http://on.wsj.com/jkZPs9>

6 May via web ☆ Favorite ↺ Retweet ↻ Reply

Twitter

	Current	1 Year History		
		Trend	High	Low
All Departments	\$ 159.6		\$ 159.6	\$ 130.5
Women's Apparel	\$ 32.6	↑	\$ 32.6	\$ 30.2
Patio & Garden	\$ 16.2	→	\$ 16.9	\$ 15.1
Toys	\$ 15.7	↑	\$ 15.7	\$ 13.5
Electronics	\$ 14.9	↓	\$ 18.3	\$ 14.9
Baby Apparel	\$ 14.2	→	\$ 14.8	\$ 7.5
Men's Apparel	\$ 13.5	→	\$ 13.2	\$ 12.8
Kid's Apparel	\$ 12.6	→	\$ 12.9	\$ 11.9
Furniture	\$ 11.4	→	\$ 14.5	\$ 8.7
Bed & Bath	\$ 11.1	↑	\$ 11.1	\$ 9.3
Kitchen	\$ 9.9	↑	\$ 10.1	\$ 7.3
Home	\$ 7.5	→		

Microsoft Excel

Chapter
2.2

Metaphors and Symbols

Metaphor



I am a walking encyclopedia.

11

A **metaphor** is a figure of speech that, for rhetorical effect, directly refers to one thing by mentioning another.^[1] It may provide clarity or identify hidden similarities between two ideas.

[Wikipedia, Metaphor]

*Die **Metapher** ist eine rhetorische Figur, bei der ein Wort nicht in seiner wörtlichen, sondern in einer übertragenen Bedeutung gebraucht wird, und zwar so, dass zwischen der wörtlich bezeichneten Sache und der übertragen gemeinten eine Beziehung der Ähnlichkeit besteht.*

[Wikipedia, Metapher]

Brainstorming

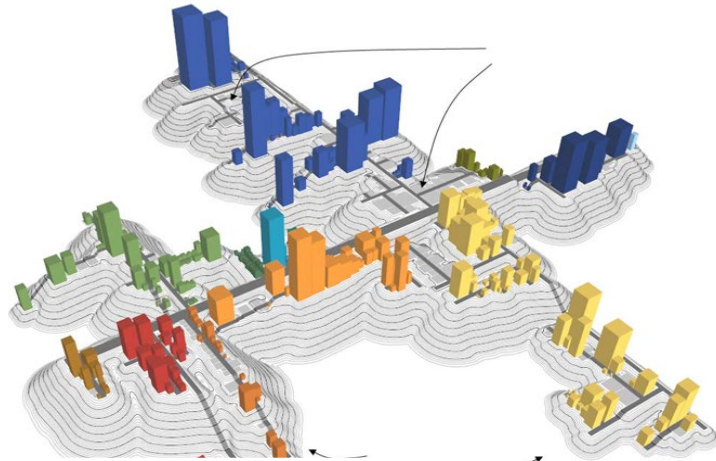


12

*Metaphors
in computer science?*

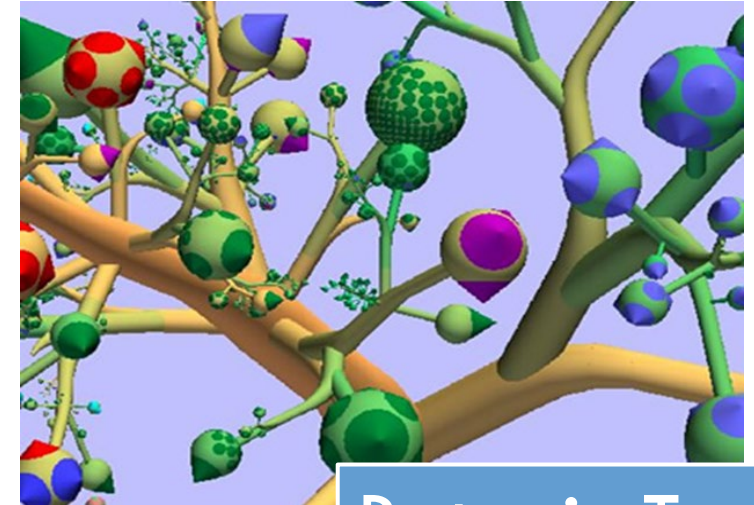
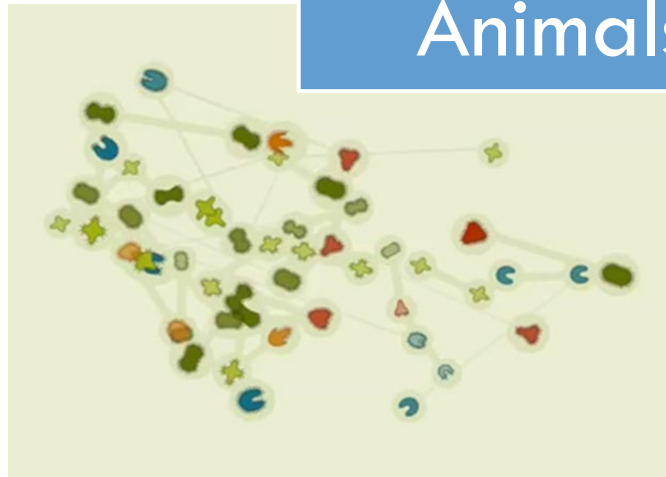
Metaphors in Visualization

13



Cities

Animals



Botanic Trees

here: microbes

<http://carohorn.de/anymails/>

Symbols/Pictograms

14

- Highly simplified pictorial representation of objects or activities.



Olympic Games London 2012

- Very suitable for depicting metaphors



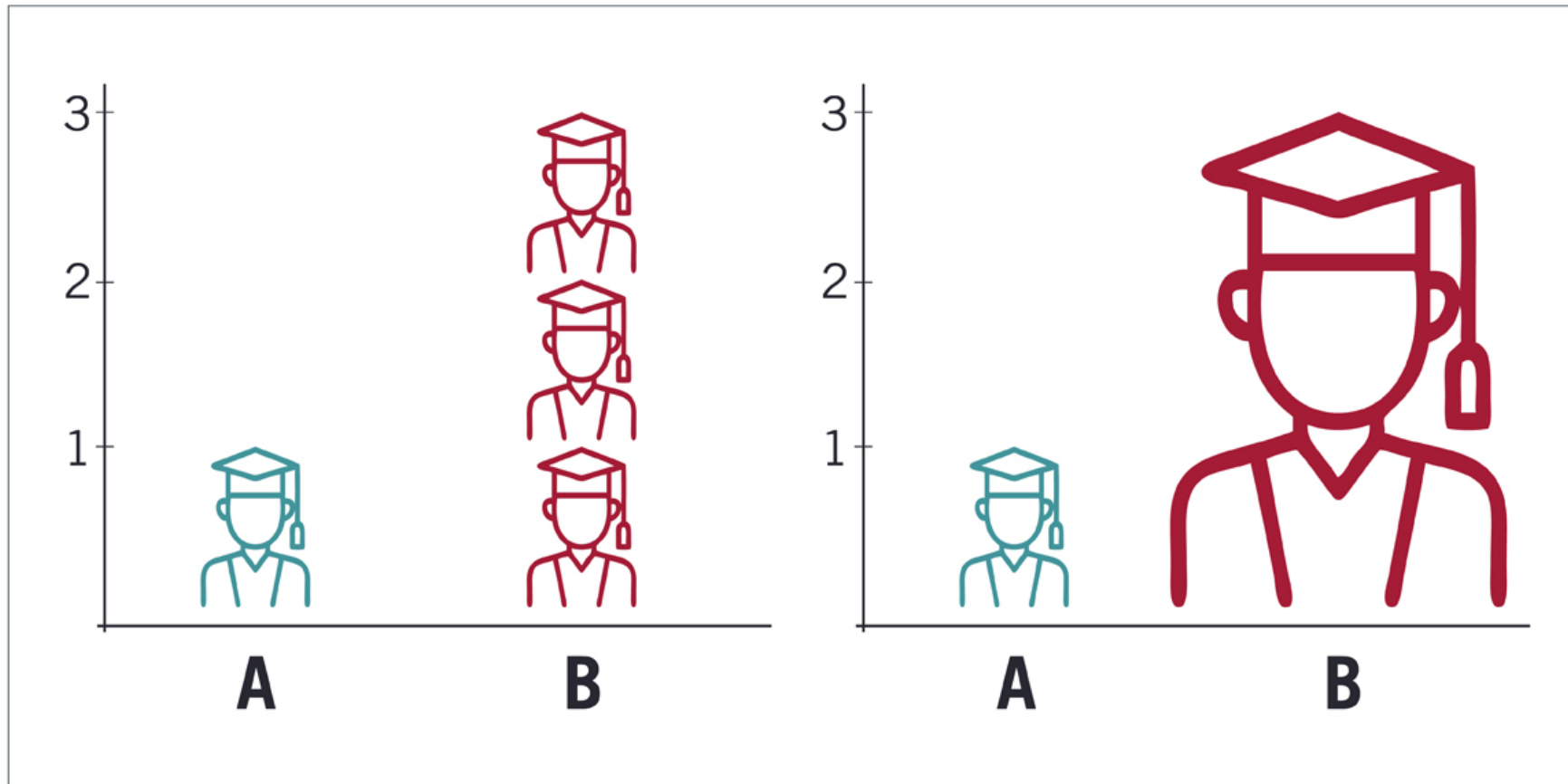
ISOTYPE

- Pictorial language „Isotype“ or „Wiener Methode der Bildstatistik“, Otto Neurath, Gerd Arntz, **1925-1934**
- **Using pictograms to convey statistical information**
- Quantity is **better** represented by the **number** of pictograms **than** by the **size** of a pictogram.



Size vs Number of Pictograms

16



Area increases quadratically
 $A = \text{width} * \text{height}$
if width and height are increased at the same time.



Gerd Arntz designed around 4000 signs, which symbolized key data from industry, demographics, politics and economy, for the visual language Isotype. [read more](#)

Isotype

people
at work

Statistics

nature
food

Activism

products
mobility

Gerd Arntz

industry
architecture

More info

various

gerd
arntz

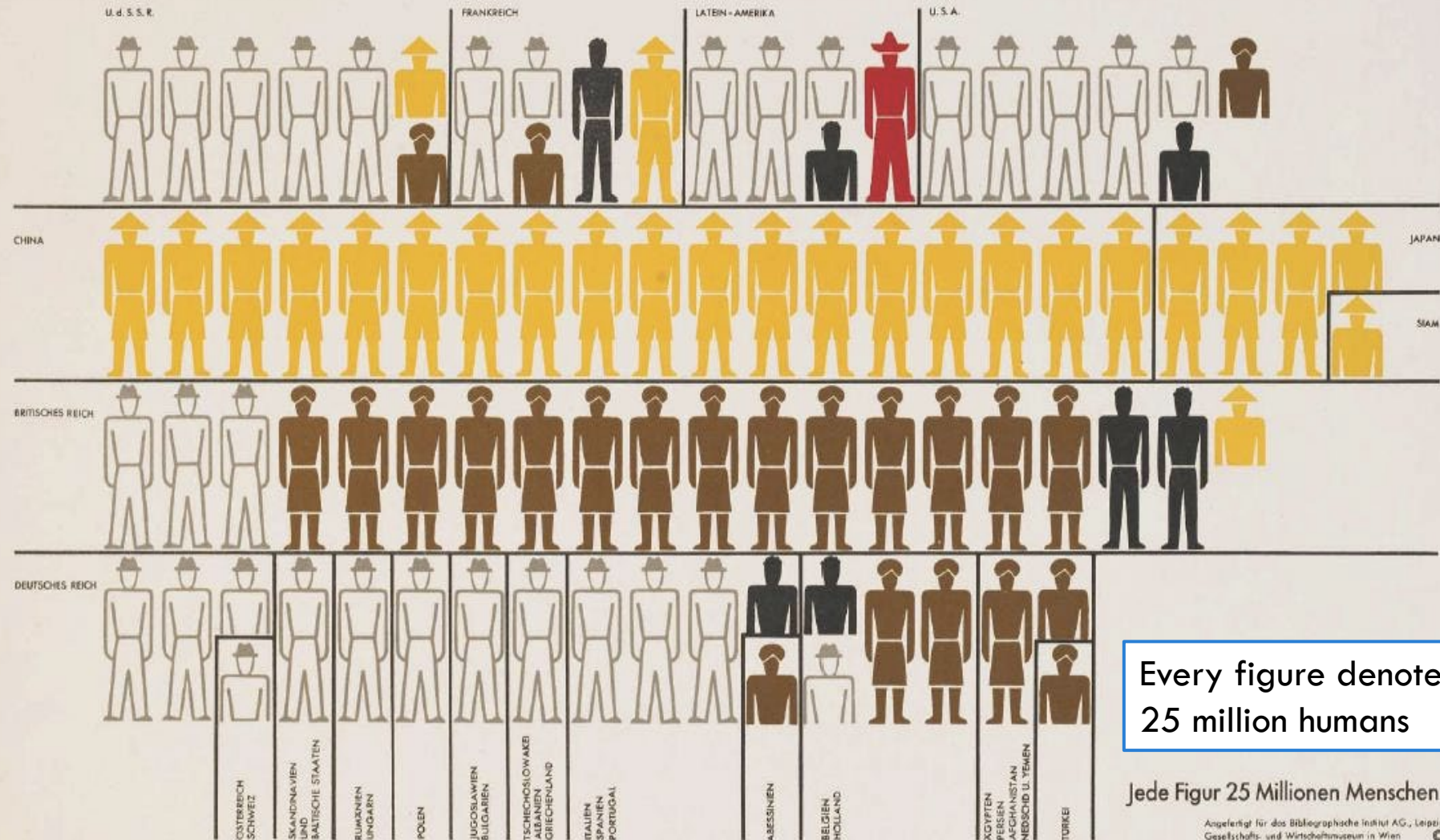
web
archive



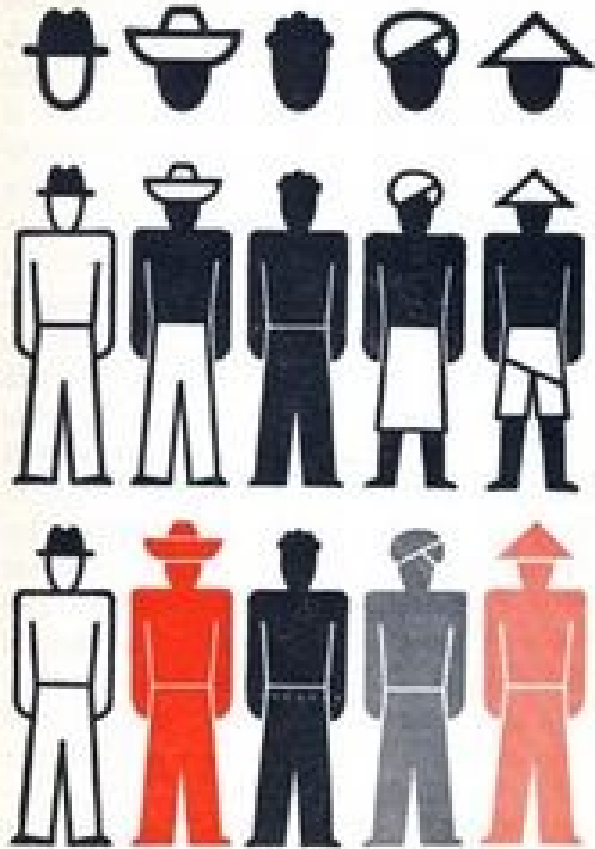
free app

World powers

Mächte der Erde



Signs for the 5 groups of men



Picture 15

Children of Their Time

While some of these symbols may reflect outdated or racially charged worldviews, it's important to view them in context. Two of the three creators of the Isotype system fled Nazi Germany and found refuge in Britain. The third, Arntz, was reportedly a committed communist:

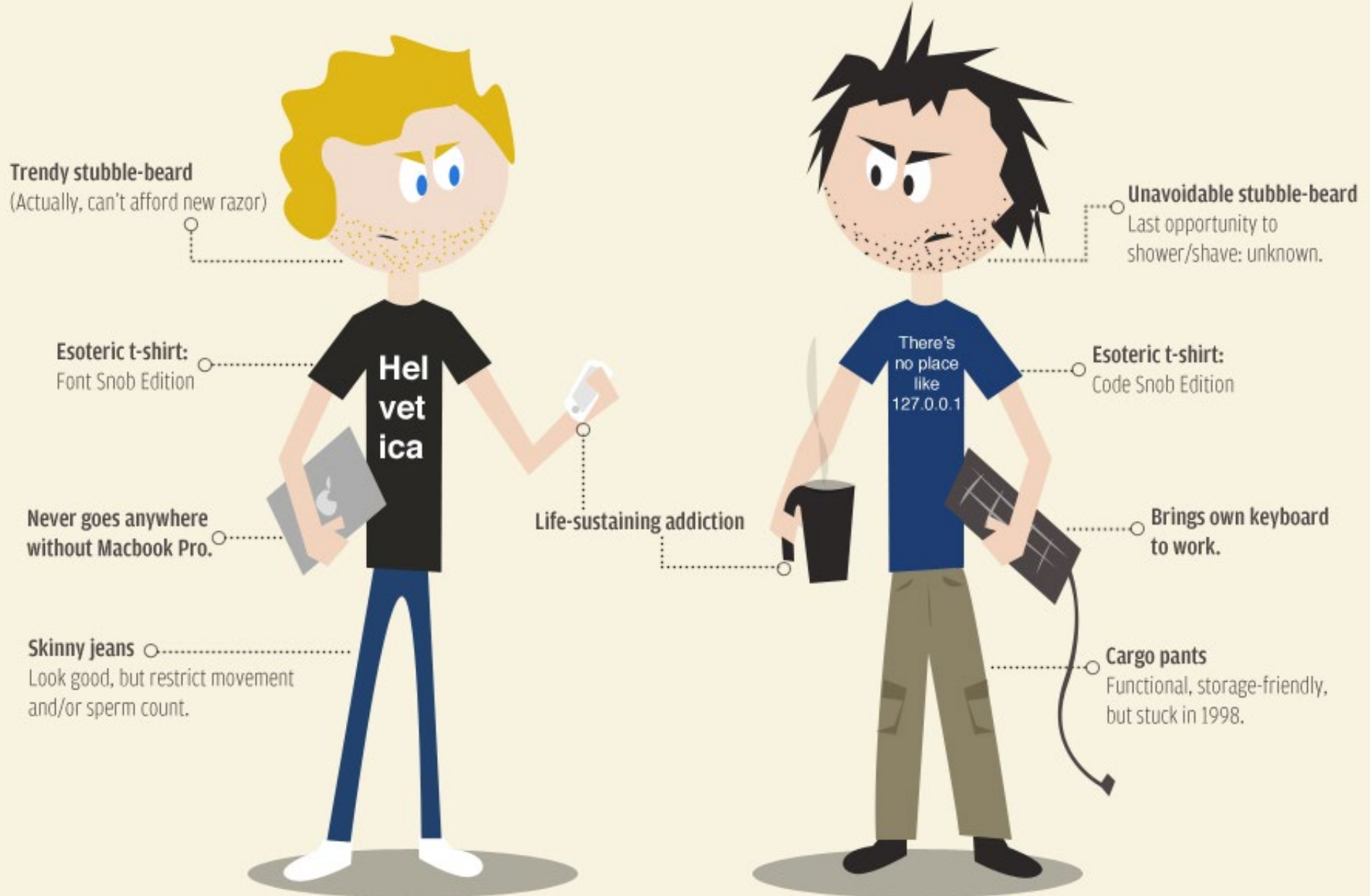
<http://isotyperevisited.org/2012/08/introduction.html>

Chapter
2.3

Infographics

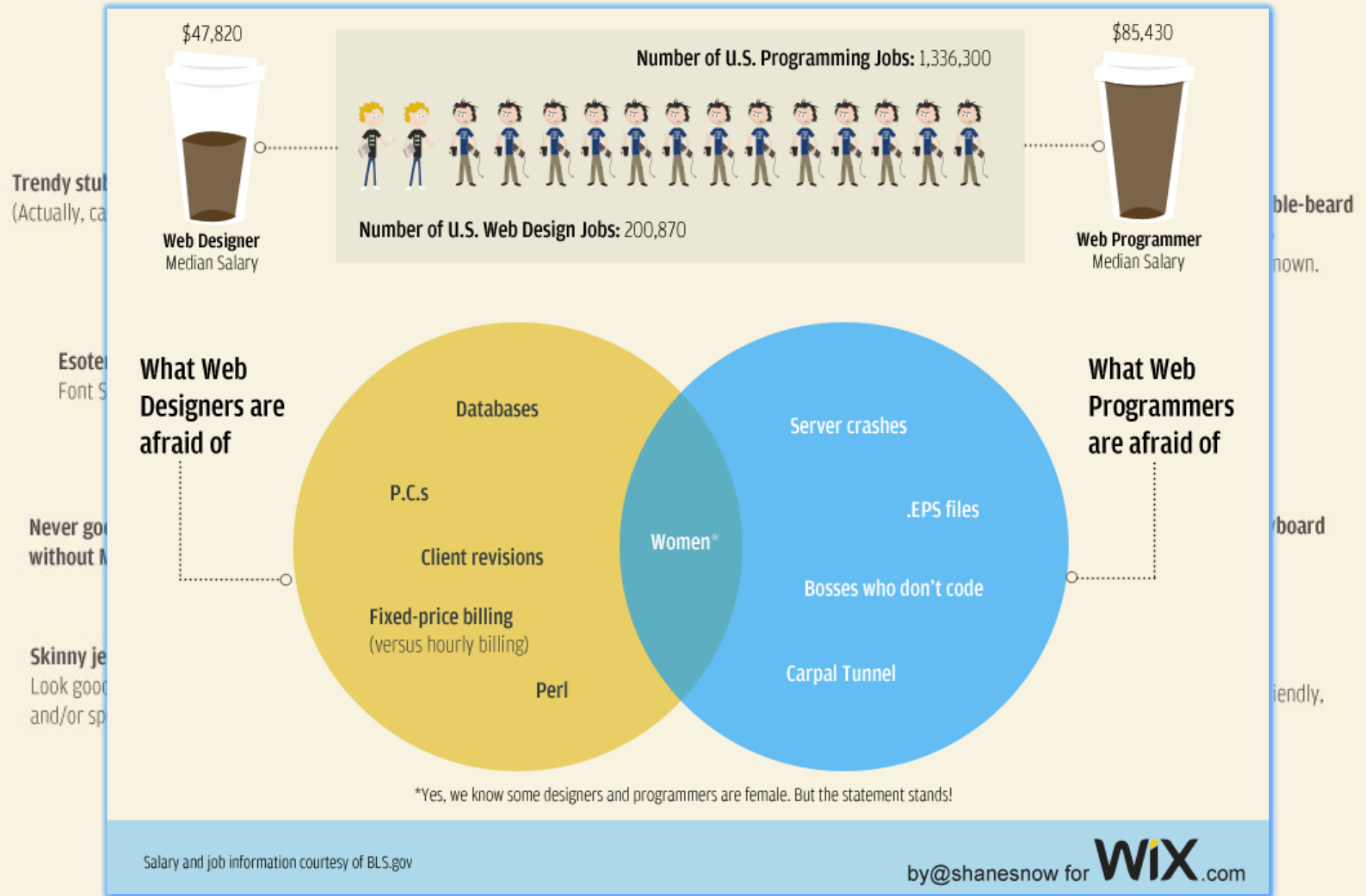
WEB DESIGNERS VS WEB DEVELOPERS

21



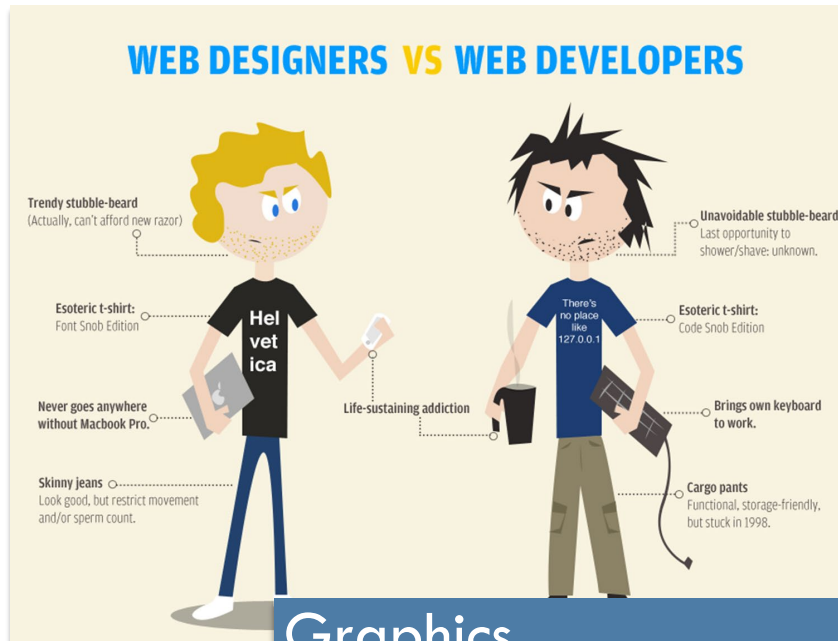
WEB DESIGNERS VS WEB DEVELOPERS

22



Elements of an infographics

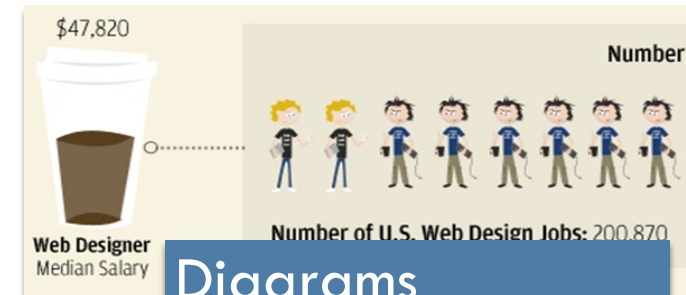
23



Graphics

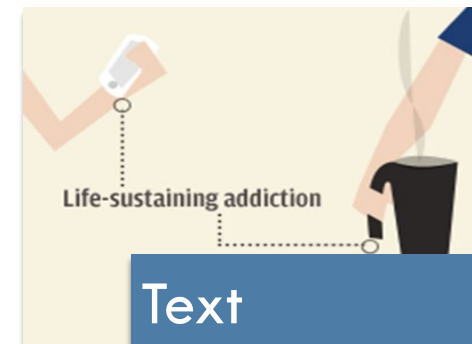
- illustrative
- simplified

Story



Diagrams

- Connected to graphics



Text

- keywords and short texts

Definition

24

Information graphics or infographics are graphic visual representations of information, data or knowledge. These graphics present complex information quickly and clearly, such as in signs, maps, journalism, technical writing, and education.

[Information Graphics, Wikipedia]

Infographics vs Information Visualization

25

Infographics

- ❑ Manually created
- ❑ Specially designed for a particular data set
- ❑ self-explanatory

Information Visualization

- ❑ Automatically computed
- ❑ Suitable for a variety of data sets
- ❑ not necessarily self-explanatory

<http://eagereyes.org/blog/2010/the-difference-between-infographics-and-visualization>

Interactive Infographics

26

Wall of Debt
World Cup planner
Health of England
Battle of Britain

iPad graphics by Applied Works

Application Areas

27

News

- newspapers
- TV
- internet



- ## Software systems
- help systems / tutorials
 - data visualizations
 - games



Presentations

- slides
- posters



Learning material

- text books
- web sites
- manuals

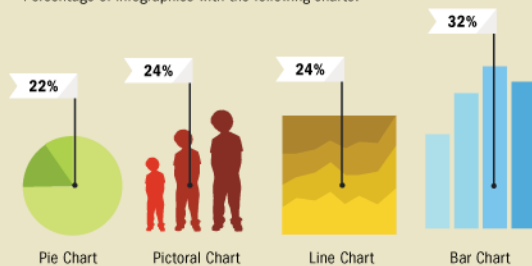
INFOGRAPHIC OF INFOGRAPHICS

Data visualization is a popular new way of sharing research. Here is a look at some of the visual devices, informational elements, and general trends found in the modern day infographic.

DESIGN

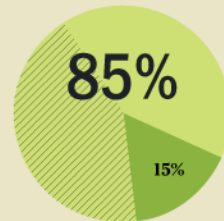
CHART STYLE

Percentage of infographics with the following charts:



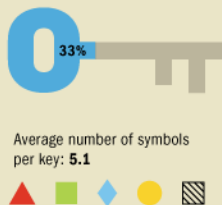
FONT

■ Sans Serif
 ▨ Condensed Sans Serif
 ■ Serif

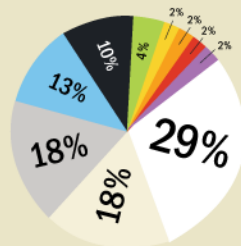


KEY INFO

Percentage of infographics with key:

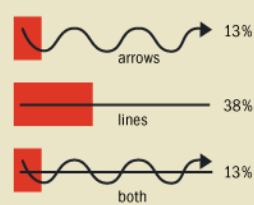


BASE COLOR



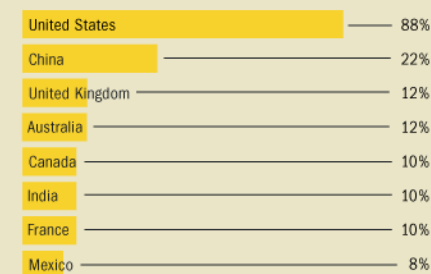
NAVIGATIONAL ICONOGRAPHY

Frequency of arrows & connecting lines in infographics:



CONTENT

COUNTRIES FEATURED



THEME

Relative popularity of different infographic themes:



SECTIONS



CREDITED SOURCES

Average number of sources per infographic: 2.29



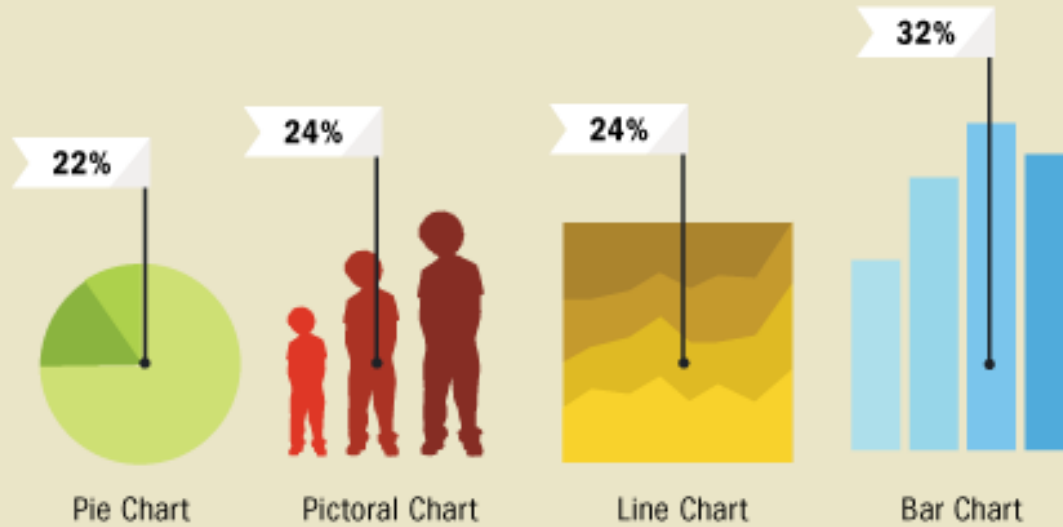
TITLE

Average number of words per infographic title: 4.36

“RICHEST AND POOREST AMERICAN NEIGH

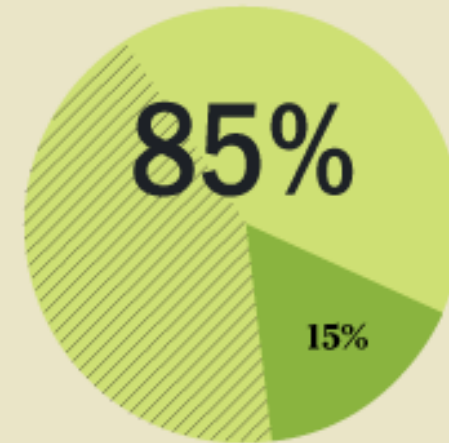
CHART STYLE

Percentage of infographics with the following charts:



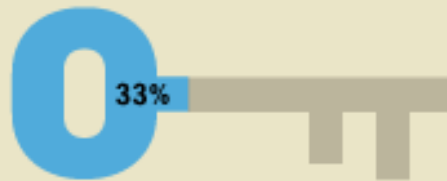
FONT

Sans Serif Condensed Sans Serif
Serif



KEY INFO

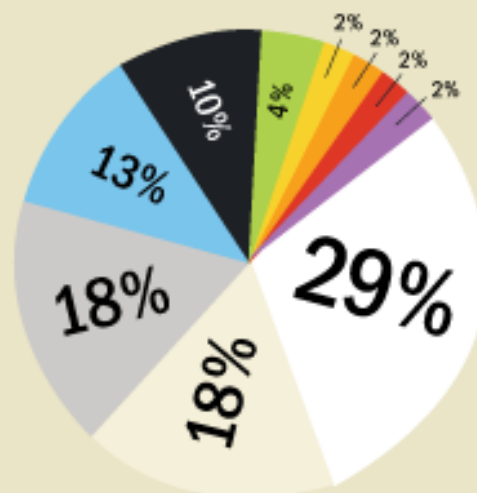
Percentage of infographics with key:



Average number of symbols per key: **5.1**

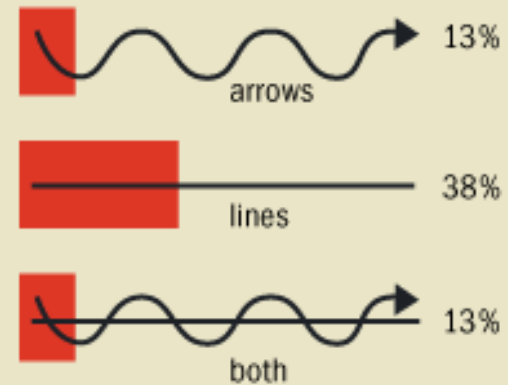


BASE COLOR



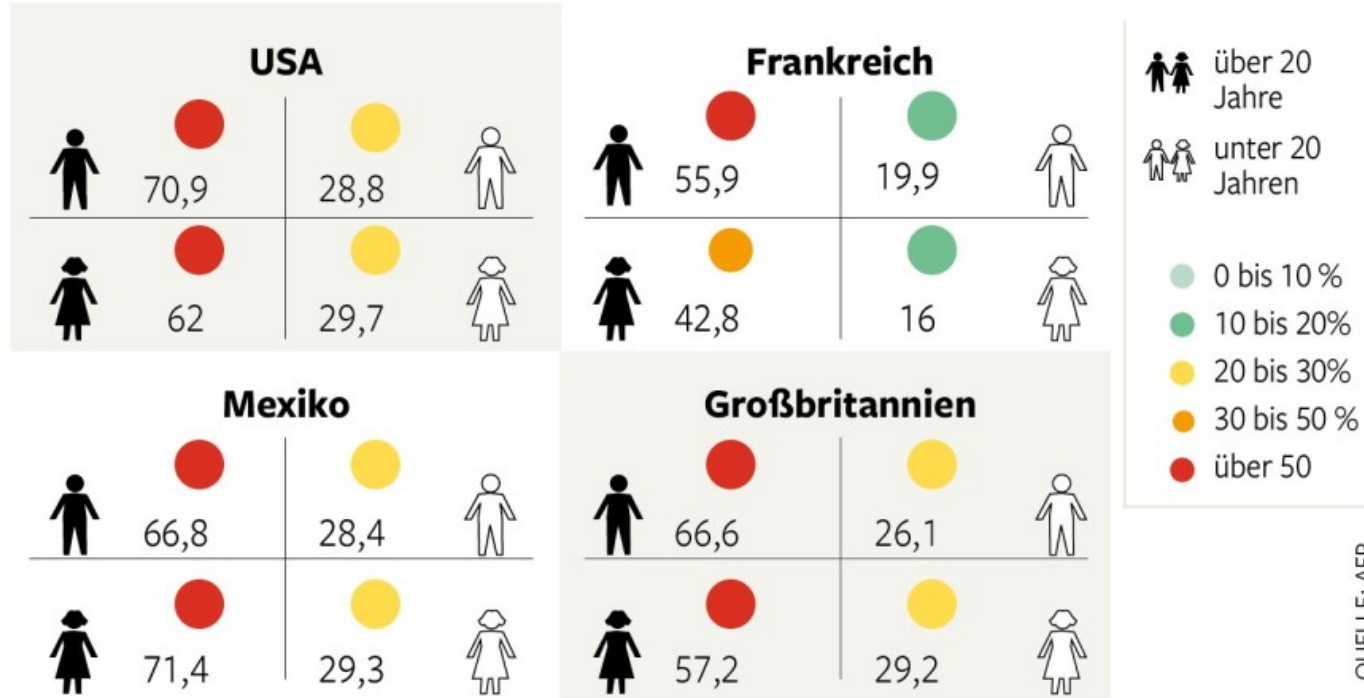
NAVIGATIONAL ICONOGRAPHY

Frequency of arrows & connecting lines in infographics:



ÜBERGEWICHTIGE UND FETTLEIBIGE

in ausgewählten Ländern (BMI > 25), Anteil an der jeweiligen Gesamtbevölkerung



QUELLE: AFP

DIE WELT

Fettleibigkeit im Ländervergleich

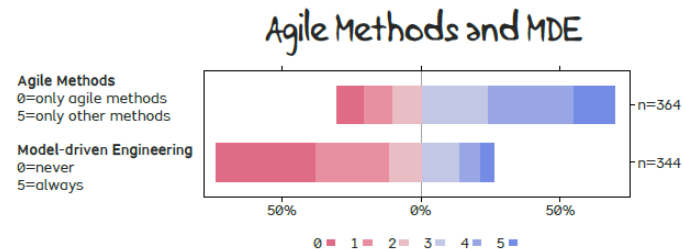
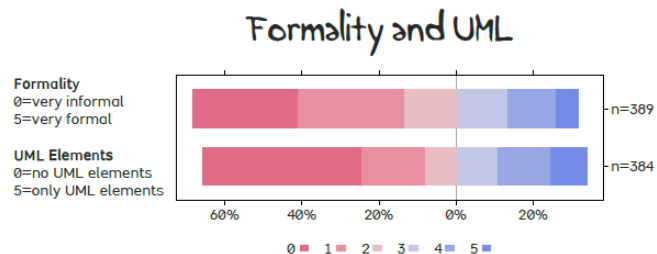
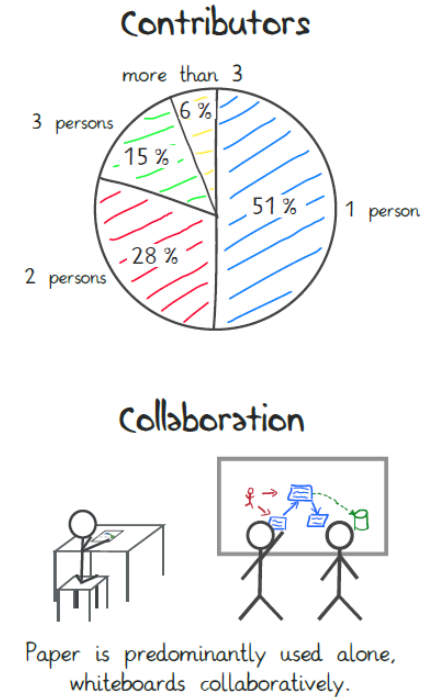
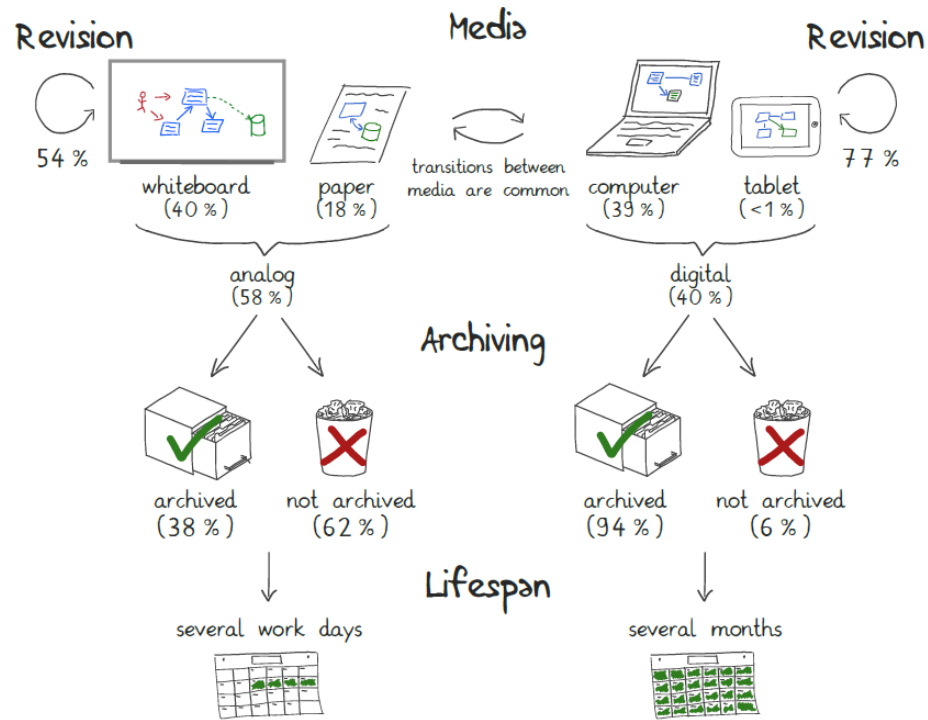
Anteil an der Bevölkerung mit einem BMI größer als 30 im Jahr 2010



ZEIT-Grafik/Quelle: OECD Health Data 2010

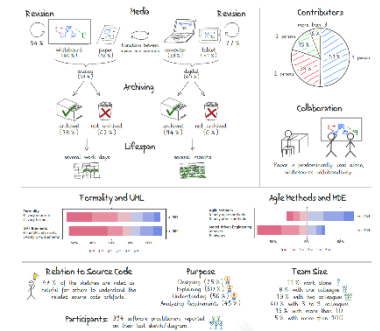
Sketches and Diagrams in Practice

How and why do software practitioners use sketches and diagrams in their daily work?



Sketches and Diagrams in Practice

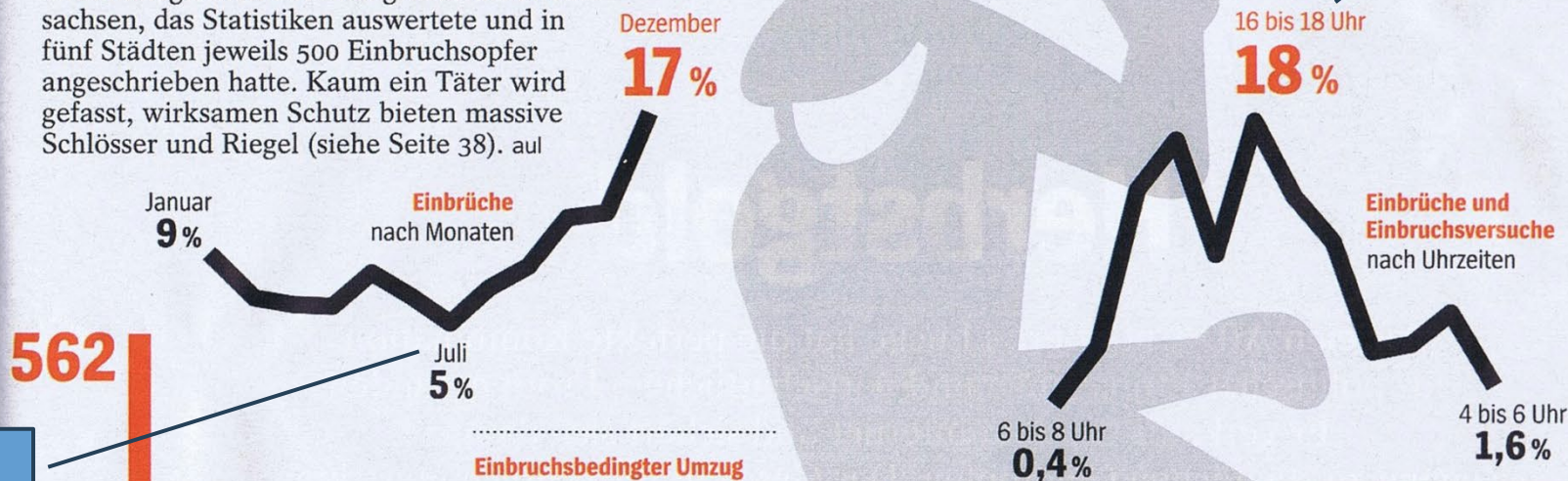
How and why do software practitioners use sketches and diagrams in their daily work?



Einbrecher Sie kommen am Tag

Mit der dunklen Jahreszeit steigt die Zahl der Wohnungseinbrüche. Die meisten Einbrecher kommen tagsüber, Einfamilienhäuser und Erdgeschosswohnungen sind am stärksten gefährdet. Das ergab eine Untersuchung des Kriminologischen Forschungsinstituts Niedersachsen, das Statistiken auswertete und in fünf Städten jeweils 500 Einbruchsoffer angeschrieben hatte. Kaum ein Täter wird gefasst, wirksamen Schutz bieten massive Schlösser und Riegel (siehe Seite 38). auf

Most burglaries happen during the day.



Burglaries are least likely to happen during the summer holidays.



In den untersuchten Städten werden pro 100 Fälle zwischen **zwei und fünf Personen** wegen eines Einbruchs verurteilt.

Einbruchsbedingter Umzug

10 Prozent der Einbruchsoffer sind wegen des Einbruchs umgezogen. Weitere 15 Prozent hätten wegen des Einbruchs umziehen wollen.

Methoden der Diebe

Art und Weise des Einbruchs in

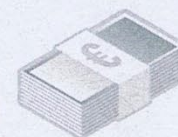
■ Mehrfamilienhäuser
■ Einfamilienhäuser

Glasscheibe einschlagen

Gefährliche Etage

37 Prozent der Einbruchsoffer in Mehrfamilienhäusern wohnten im Erdgeschoss.

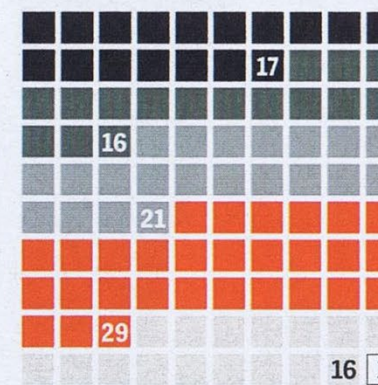
Quelle: Kriminologisches Forschungsinstitut, 2014



Schaden

Gesamtwert der gestohlenen Objekte, in Prozent der Fälle

■ über 10 000 €
■ 5001 bis 10 000 €
■ 2001 bis 5000 €
■ 501 bis 2000 €
■ 50 bis 500 €
□ unter 50 €



Wer bekommt die 100,- Euro für meine Sportschuhe?



gefördert durch die:

Österreichische
Entwicklungszusammenarbeit



Dieser Poster wurde mit finanzieller Unterstützung der Europäischen Gemeinschaft erstellt. Die darin enthaltenen Standpunkte geben die Ansicht der Slowakei-Agentur wieder und stellen nicht in keiner Weise die offizielle Meinung der Europäischen Gemeinschaft dar.

CLEAN CLOTHES
kampagne für faire arbeitsbedingungen weltweit
www.cleanclothes.at

Energiequelle	Energieträger	Antrieb	Lokal emissionsfrei	Eine 3-MW-Windkraftanlage versorgt...
Strom  z.B. Windkraftanlage 3 Megawatt Leistung, 2000 Stunden Vollast pro Jahr	Strom	 BEV		 1600 Fahrzeuge
	Wasserstoff	 FCEV		 600 Fahrzeuge
	eFuel	 ICE		 250 Fahrzeuge

Energy Source

Energy Carrier

Drive System

Local
Emissions-Free

A 3-MW Wind Turbine
Powers...

Electricity



e.g. Wind
Turbine

3 Megawatts
Capacity,
2000 Full-Load
Hours per Year

Electricity



BEV

(Battery Electric Vehicle)



1600
Vehicles

Hydrogen



FCEV

(Fuel Cell Electric Vehicle)



600
Vehicles

eFuel



ICE

(Internal Combustion Engine)



250
Vehicles

More examples

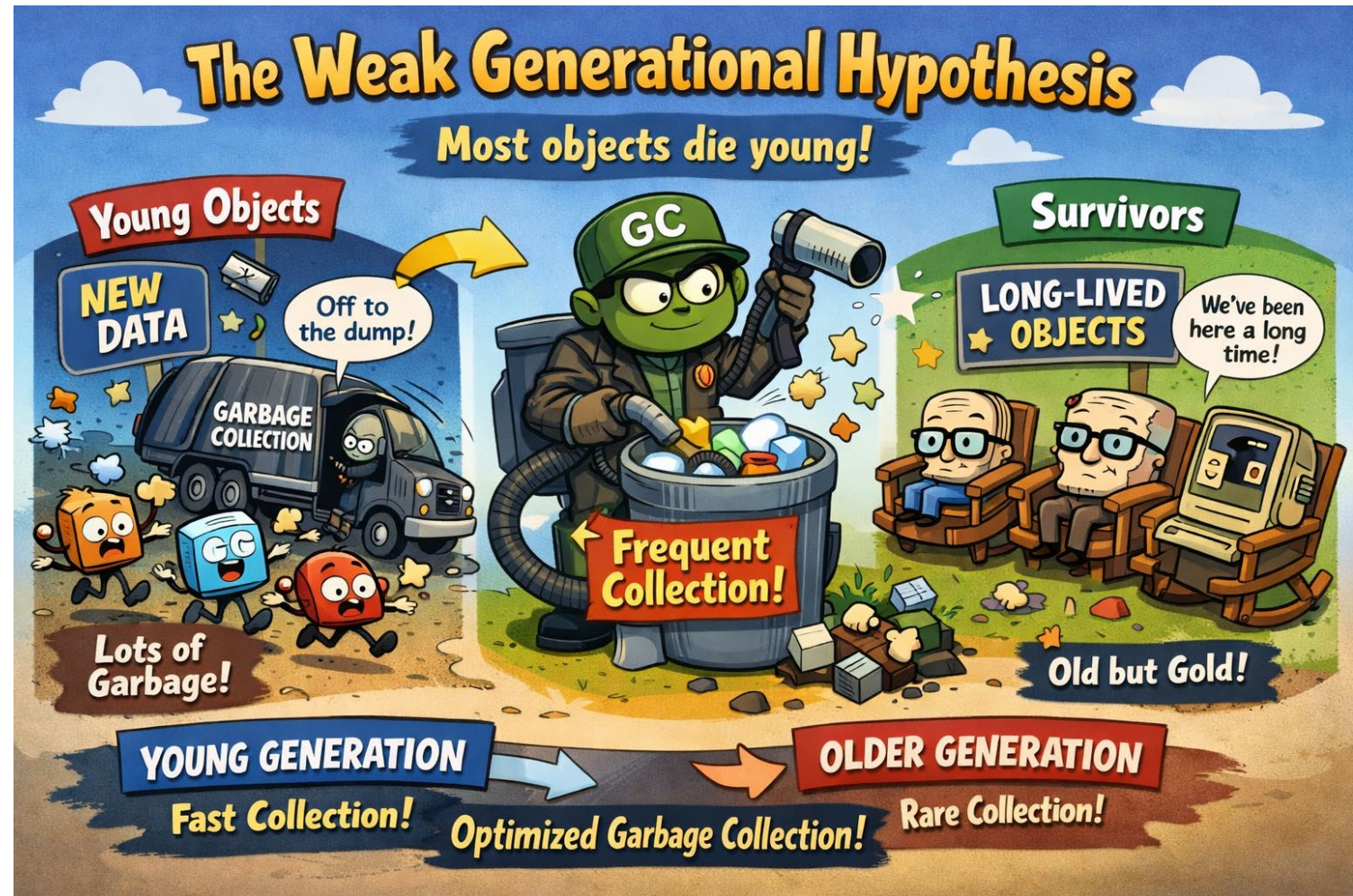
36

- <https://www.siegemedia.com/creation/best-infographics>

Generating Infographics with AI

37

Generate Infographics based on description of „Weak Generational Hypothesis“ in Wikipedia.



Generating Infographics with AI

38

Ask AI to animate image.



Exercise

Task:

Design an original and informative infographic related to a topic of your own choice. Try to make it self-explanatory and appealing.

Software for creating infographics

40

Vector graphics

- **Vektor graphics tools:** Inkscape, Corel Draw, Adobe Illustrator
- **Layout software:** Adobe InDesign, Microsoft Publisher, Scribus
- **Presentation programs:** Microsoft Power Point, OO/LO Impress, Apple Keynote

Diagrams

- **Diagraming tools:** Microsoft Visio, yEd, Creatly, Dia
- **Spread sheets:** Microsoft Excel, OO/LO Calc
- **Scripts:** gnuplot, R

Interactive Visualizations

- **Tools:** Tableau
- **Frameworks:** JFreeChart (Java), Google Chart Tools (JavaScript), Processing (Java)

underlined: free of charge

Q-map tool for infographics (f.e. used by NZZ): <https://github.com/nzzdev/Q-map>

Suggestions for topics

41

Guiding questions:

- ☐ *What information is relevant?*
- ☐ *Are data sources for this information available?*
- ☐ *How can the information be visually represented?*
- ☐ *What are suitable metaphors?*
- ☐ *What could be the story of your infographic?*



Renewable
Energy



Demographics



Mass Sport
(Breitensport)



Alternative
Medicine



Cybercrime

First Tutorial

42

- Tuesday, **28.04.2026**, 10-12 in HZ 203
 - 1st and 2nd exercise sheet